Prop 2.3.14 $f: X \to Y$ morph of var. $\forall \mathcal{F} \in D^b_c(Y, \Bbbk)$. $f^*\mathcal{F} \in D^b_c(X, \Bbbk)$

2.3.15 $i: Y \hookrightarrow X$ loc closed subvar. $\forall \mathcal{F} \in D^b_c(Y, \Bbbk)$ $i_*\mathcal{F} \in D^b_c(X, \Bbbk)$

2.3.17 $\mathcal{F}, \mathcal{G} \in D^b_c(X, \Bbbk)$ $\mathcal{F} \otimes \mathcal{G}$ & $R\mathcal{H}om(\mathcal{F}, \mathcal{G}) \in D^b_c(X, \Bbbk)$

2.3.18 $\in D^b_c$ $\mathcal{F} \otimes \mathcal{G} \in D^b_c$

2.3.19 f: finite morph. $\mathcal{F} \in D^b_c \Rightarrow f_*\mathcal{F} \in D^b_c$

Y closed $\Rightarrow i_*\mathcal{F} \in D^b_c(X, \Bbbk)$

2.4.2

X smooth var. Z div with s.n.c.

$j_I: X_I \hookrightarrow X$ inclusion of stratum in the normal crossing divisor $\perp$ $\mathcal{L}_I$ on X_I constr. finite type

$\Rightarrow j_{I*}\mathcal{L}$

coh supp $H^i(j_*\mathcal{L}) \supset |I| + i$

- Sheaf functor & constructibility

$f: X \to Y$ morph of var.

$\mathcal{F} \in D^b_c(X, \Bbbk)$

$f_*\mathcal{F}$ & $f^!\mathcal{F} \in D^b_c(Y, \Bbbk)$

Induction on $\dim X$.

i: open embedding

p proper.

Nagata $X \xrightarrow{j} \bar{X} \xrightarrow{p} Y$

open $f_*\mathcal{F}$ ✓.

Prop: $f_*\mathcal{F} = f_*\mathcal{F}$

$$\begin{tikzcd} p^{-1}(U) \arrow[r, "\bar{j}"] \arrow[d, "p"] & \bar{X} \arrow[d, "p"] \\ U \arrow[r, hook, "j"] & X \end{tikzcd}$$

$\bar{j}$ open dense

$Z = \bar{X} \setminus X$ closed subset

$f_*\mathcal{F}|_U = \bar{j}^*\bar{j}_*\mathcal{F}$

$\bar{j}_*\mathcal{F}$ constr. $\Rightarrow p_*\bar{j}_*\mathcal{F}$

open $\cup \{f_*\mathcal{F}|_U\}_{U \text{ open}}$ constr.

enough to

$i^*f_*\mathcal{F} = i^*p_*\bar{j}_*\mathcal{F}$

$\cong p_*i^*\bar{j}_*\mathcal{F}$

$\cong p_*(\underbrace{\bar{j}_*\mathcal{F}|_Z}_{\mathcal{F}})$

constr. 2.4.2 good stratification

& p_*: induction $\dim Z < \dim U + \dim X$

- Truncation

$\Rightarrow \mathcal{F}$ constructible sheaf.

X irr.

$i: X \hookrightarrow X$ inclusion of an irr. comp.

$j: U \hookrightarrow X$ comp.

$f_* = j_!j^*\mathcal{F} \to f_*\mathcal{F} \to f_*(\mathcal{F}|_Z) = i_*i^*\mathcal{F}$

$$\begin{tikzcd} U \arrow[r, hook, "j"] \arrow[dr, hook, "j'"'] & X \arrow[r, "f"] & Y \\ & \bar{U} \arrow[u, hook] \arrow[ur, "f'"'] & \end{tikzcd}$$

$f_*j_{!!} = f'_*j_{!!}$

& Both f_* & $(f_*i)_*$ are of finite irr comp.

$U \hookrightarrow X \xrightarrow{j} W$

$\downarrow f$

Y

$\Rightarrow h_*\mathcal{L}$ constr.

$(f_*j)_*i^*\mathcal{F} \to f_*f^*\mathcal{F} \to f_*h_*\mathcal{L} \to$.

Reduction:

$f_*h_*\mathcal{L}$ constr.

$i_*i^*\mathcal{F} \to \mathcal{F} \to h_*\mathcal{L}$

- $\mathcal{F}$ constr.
- $h_*\mathcal{L}$ constr by Step 2 $\Rightarrow i_*i^*\mathcal{F}$ constr.

$i: Z \hookrightarrow X$

$\Rightarrow i^*\mathcal{F}$ constr.

$\dim Z < \dim X \Rightarrow (f_*i)_*$ preserves constr. by ind.

induction on $\dim Y$

$\underbrace{(f_*i)_*i^*\mathcal{F}}_{\mathcal{F}} \to f_*\mathcal{F} \to f_*h_*\mathcal{L} \to$

- preserves
- to prove $f_*\mathcal{F}$ constr. only need to prove $f_*h_*\mathcal{L}$ constr.

- $\dim Y = 0 \Rightarrow R\Gamma(\mathcal{L})$ constr by Artin vanishing thm

$$\begin{tikzcd} p^{-1}(U) \arrow[r, hook] \arrow[d, equal] & \bar{X} \arrow[r, hook] \arrow[d, "p"] & p^{-1}(Z) \arrow[d] \\ U \arrow[r, hook, "h"] & X \arrow[r, hook] \arrow[d, "f"] & Z \\ & Y & \end{tikzcd}$$

$\mathcal{F}^{-1}(U) \hookrightarrow \bar{X} \hookrightarrow p^{-1}(Z)$

$U \hookrightarrow X \hookleftarrow Z$

$\mathcal{F}^{-1}(U) \hookrightarrow \bar{X} \hookleftarrow p^{-1}(Z)$

- $\dim Y > 0 \Rightarrow f$ proper, p: $\bar{X} \to X$

$\mathcal{F}^{-1}(U) \hookrightarrow \bar{X} \hookleftarrow p^{-1}(Z)$

$\mathcal{F}^{-1}(U) \hookrightarrow \bar{X} \hookleftarrow p^{-1}(Z)$

$\mathcal{F}_1 := \mathcal{F}_2$ irr comp of $\mathcal{F}$

2.1.8 $\Rightarrow \exists V \subseteq Y$ st.

$\exists f^{-1}(V) \cap \mathcal{F}_2$: $\bar{f}^{-1}(V) \cap \mathcal{F}_2 \to V$

smooth & proper.

$\Rightarrow \bar{f}^{-1}(V) \to V$ surj

- $\bar{f}: \bar{U} \to V$ transverse to

$$\begin{tikzcd} \mathcal{F}^{-1}(U) \arrow[r, hook] \arrow[d, equal] & \bar{X} \arrow[r, hook] \arrow[d, "p"] & p^{-1}(Z) := \mathcal{F} \\ U \arrow[r, hook, "h"] & X \arrow[r, hook] \arrow[d, "f"] & Z \\ \mathcal{F} & Y & \end{tikzcd}$$

$\bar{f}^{-1}(V) := V'$

$\cup$

$\cap \mathcal{F}$

$\Rightarrow$ transv. locally trivial

$$\begin{tikzcd} V' \arrow[d, "\bar{f}'"] \\ V \end{tikzcd}$$

$j^*(f_*h_*\mathcal{L}) \cong j^*(\bar{f}_*\bar{h}_*\mathcal{L})$

$\cong \bar{f}'_*\, j^*(\bar{h}_*\mathcal{L})$

$\cong \bar{f}'_*\, j^*(\underbrace{h_*\mathcal{L}}_{\text{constr.}})$ $\xleftarrow{\text{d-lyc}}$

2.4.3 $f_*(R_*\mathcal{L})|_Q \cong (a_{\bar{f}^{-1}(y)})_* (R_*\mathcal{L})|_{\bar{f}^{-1}(y)}$

constr.: for $\forall$ inherits this struct

constr.: stalk at any pt. $y \in V$ is const.

$\bar{f}$ smooth $\Rightarrow$ $\dim \bar{f}^{-1}(y)$

$= \dim X - \dim Y < \dim X$

$f: X \to Y$ smooth morph. of smooth var.

$Z \subset X$ div with snc f transv. locally triv. fib.

$\mathcal{F} \in D^b_c(X, \Bbbk)$ weakly constr. w.r.t. $\mathcal{L}$

$\Rightarrow f_*\mathcal{F} \in D^b_{lc}(Y, \Bbbk)$

moreover $(f_*\mathcal{F})_y \cong R\Gamma(\mathcal{F}|_{f^{-1}(y)})$ isom.

Thm 2.7.3. $f: X \to Y$ morph

$\mathcal{F} \in D^b_c(X, \Bbbk)$ $f_*\mathcal{F}$ $f_!\mathcal{F} \in D^b_c(Y, \Bbbk)$

Lem 2.7.2. $i: Z \hookrightarrow X$ closed emb.

$\mathcal{F} \in D^b_c(X, \Bbbk)$. $\Rightarrow i^*\mathcal{F} \in D^b_c(Z, \Bbbk)$

$j: U \hookrightarrow X \hookleftarrow Z: i$ open

$i_*i^*\mathcal{F} \to \mathcal{F} \to j_*j^*\mathcal{F} \to$.

$\mathcal{F}$ constr. ✓ $j^*\mathcal{F}$ constr. triv. $\Rightarrow j_*j^*\mathcal{F}$ constr. 2.7.2

In The Proof of 2.7.1 !

$\Rightarrow i_*i^*\mathcal{F}$ constr.

$\Rightarrow i^! = i^*(i_*i^*\mathcal{F})$

Prop 2.7.3: $\mathcal{F}, \mathcal{G} \in D^b_c(X, \Bbbk)$

$\Rightarrow R\mathcal{H}om(\mathcal{F}, \mathcal{G}) \in D^b_c(X, \Bbbk)$.

$U \xrightarrow{j} X \hookleftarrow Z$.

$i_*R\mathcal{H}om(\underbrace{i^*\mathcal{F}}_{\text{constr. triv.}}, \underbrace{i^!\mathcal{G}}_{\text{constr.}}) \to R\mathcal{H}om(\mathcal{F}, \mathcal{G}) \to j_*R\mathcal{H}om(\mathcal{F}|_U, \mathcal{G}|_U) \to$.

Exer 1.3.7.

Like $i_*i^!\mathcal{F} \to \mathcal{F} \to j_*j^*\mathcal{F}$

$\Rightarrow R\mathcal{H}om(i^*\mathcal{F}, i^!\mathcal{G})$ constr. $\Rightarrow i_*R\mathcal{H}om(i^*\mathcal{F}, i^!\mathcal{G})$ const

& $R\mathcal{H}om(\mathcal{F}|_U, \mathcal{G}|_U)$ 2.3.27

$\Rightarrow \square$.

Cohomology vanishing results.

Thm 2.7.4: X var of dim n $\mathcal{F}$ constr. sheaf on X,

$\Rightarrow$ k-coef $H^k_c(X, \mathcal{F})$ f.g. $\forall k$ & vanishes unless $0 \le k \le 2n$.

$a_X: X \to pt$

$a_{X*}\mathcal{F}$ constr.

$R\Gamma_c$ ○ $f_!f^*\mathcal{F} \to \mathcal{F} \to i_*i^*\mathcal{F} \to$

$R\Gamma_c(\mathcal{F}|_U) \to R\Gamma_c(\mathcal{F}) \to R\Gamma_c(\mathcal{F}|_Z) \to$

- $\dim Z \le n$ induction $\Rightarrow H^k_c(Z, \mathcal{F}|_Z) = 0$

$m \ge 2n$ $H^m_c(U, \mathcal{F}|_U)$ bigot.

0: $R\Gamma_c(\mathcal{F}|_U) \to \mathcal{F}^{-m} R\Gamma_c(\mathcal{F}|_U) \cong \underset{\mathcal{F}(U, \mathcal{F}|_U)}{\mathcal{M}(-m)}$

- $\mathrm{Hom}(G, (\mathcal{F}|_U) \cdot \mathcal{M}(-m)) =$

$\mathrm{Hom}(\mathcal{F}|_U, a_U^* \mathcal{M}(-m))$

Recall 2.3.1

$\mathrm{Hom}(\mathcal{F}|_U, a_U^* \mathcal{M}(m-n)(-n))$

$= \mathrm{Ext}_{\mathbb{C}(U, \Bbbk)}(\mathcal{F}|_U, \mathcal{M}(n))$

But $2n-m < 0$ Ext. vanishes

Thm 2.7.5 $H^k(X, \mathcal{F})$ f.g. $\forall k$ & vanishes unless $0 \le k \le 2n$.

Alternative proof:

X var. X of dim n admits a covering by $\varnothing$ most $n+1$ affine open subsets.

$\Rightarrow \mathcal{F}$ constr.

$H^k(X, \mathcal{F})$ f.g. $\forall k$ & vanishes unless $0 \le k \le 2n$

Cant take argument of 2.7.4 because

$(a_{i*}a^*)$ ✓

(a^*a_{i*}) ✗

2.4.2

X smooth var. Z div with s.n.c.

$j_I: X_I \hookrightarrow X$ inclusion of stratum in the normal crossing divisor $\perp$ $\mathcal{L}_I$ on X_I constr. finite type

$\Rightarrow j_{I*}\mathcal{L}$

coh supp $H^i(j_*\mathcal{L}) \supset |I| + i$

Step 1: Prove for all closed subsets of X

$\mathcal{F} \in D^b_c(X, \Bbbk)$ s.t. coh supp $H^i(\mathcal{F}) \supset i$ for $i \le n$

& $H^i(\mathcal{F}) = 0$ $\forall j \ge n$

$\Rightarrow H^k(X, \mathcal{F}) = 0$ $\forall k \ge 2n$

induction on # nonzero coh sheaves of $\mathcal{F}$

$\mathcal{F}^{n-1} \mathcal{G} \to \mathcal{G} \to H^n(\mathcal{G})[-n] \to$

$\Rightarrow R\Gamma(\mathcal{F}^{n-1}\mathcal{G}) \to R\Gamma(\mathcal{G}) \to R\Gamma(H^n(\mathcal{G})[-n])$

$H^k(X, \mathcal{F}^{n-1}\mathcal{G}) = 0$ $\forall k \ge 2n$ by ind

$H^k(X, H^n(\mathcal{G})[-n])$

$H^n(\mathcal{G})$ supported on a closed subset of dim at most n from assumption $\Rightarrow H^k(X, H^n(\mathcal{G})[-n]) = 0$ if $k \ge 2n+1$

2.7.5 ✓

Step 2: X proj: $R\Gamma(\mathcal{F}) \cong R\Gamma(\mathcal{F})$ ✓.

Step 3: X smooth & $\mathcal{F}$ loc sys

$j: X \hookrightarrow \bar{X}$ open dense.

2.1.18 assume $\bar{X}$ smooth $\bar{X} \setminus X$ div with simple n.c.

$j_*\mathcal{F}$ satisfies 1

Step 3: $H^k(X, \mathcal{F}) \cong H^k(\bar{X}, j_*\mathcal{F}) = 0$ $\forall k \ge 2n$

Lemma 2.4.2.

Step 4: general: ind on $\dim X$

$\mathcal{F}$ some ind on $\dim X$

X irr. ✓

$j: U \hookrightarrow X$ irr. smooth Zariski open

$i: Z \hookrightarrow X$ $j_*j^*\mathcal{F} \to \mathcal{F} \to i_*i^*\mathcal{F} \to$.

$\Rightarrow R\Gamma(j_*\mathcal{L}) \to R\Gamma(\mathcal{F}) \to R\Gamma(\mathcal{F}|_Z)$

enough to $H^k(X, j_*\mathcal{L}) = 0$ $\forall k \ge 2n$.

$U \to \bar{U} \hookrightarrow X$ open $X \setminus U$ irr.

U irr. smooth Zariski open subset s.t. $\mathcal{F}|_U$ is $\bar{j}$ factors through

$\bar{U} \hookrightarrow X$ assume for simplicity $X = \bar{U}$

$\Rightarrow X$ ✓✓

$$\begin{tikzcd} U \arrow[r, hook, "j"] \arrow[d, equal] & X \arrow[d, "p"] & Z \arrow[l, hook] \\ \mathcal{F}|_U \arrow[r, hook] & \bar{X} \arrow[r, "\bar{j}"] & p^{-1}(U) \arrow[u, hook] \end{tikzcd}$$

smooth

$\bar{j}: \bar{X} \to \bar{X}$

$\Rightarrow R\Gamma(j_*\mathcal{L}) \to R\Gamma(\bar{j}_*\mathcal{L})$ enough to prove $H^k(\bar{X}, \bar{j}_*\mathcal{L}) = 0$

$\bar{j}_*\mathcal{L}$

$\bar{j}_*\bar{j}^*\bar{j}_*\mathcal{L} \to \bar{j}_*\mathcal{L} \to \bar{j}_*\bar{j}^*\bar{j}_*\mathcal{L} \to$

$\bar{j}^*\bar{j}_*\mathcal{L} = \mathcal{L}$ ✓

$H^i(\bar{j}_*\mathcal{L}) \cong H^i(\bar{X}, \bar{j}^*\bar{j}_*\mathcal{L})$ $\forall i \ge 1$

$X \to Y$

lem 2.4.2 $\Rightarrow$

coh supp $H(\bar{X}, \bar{j}^*\bar{j}_*\mathcal{L}) \supset i$

$\forall i \ge 1$

$\Rightarrow$ coh supp $H^i(\bar{j}^*\bar{j}_*\mathcal{L}(1)) \supset i$

$\forall i \ge 1$

$\Rightarrow H^k(\bar{X}, \bar{j}^*\bar{j}_*\mathcal{L}(1)) = 0$ $\forall k \ge 2(n-1)$

$\Rightarrow H^k(\bar{X}, \bar{j}^*\bar{j}_*\mathcal{L}(1)) = 0$ $\forall k \ge 2n$

Use RS $\Rightarrow$

$R\Gamma(\bar{X}, \bar{j}_*\mathcal{L}(-1)) \to R\Gamma(\bar{X}, \bar{j}_*\mathcal{L}) \to R\Gamma(\mathcal{L})$

Step 3 $\Rightarrow H^k(U, \mathcal{L}) = 0$ $\forall k \ge 2n$.

Hence $\Rightarrow H^k(\bar{X}, \bar{j}_*\mathcal{L}(-1)) = 0$ $\forall k \ge 2n$

$\Rightarrow H^k(X, j_*\mathcal{L}) = 0$ $\forall k \ge 2n$ $\square$